



Modern cars allow you to start the engine, turn on the AC and control many aspects of the car from your smartphone

that, "If somehow we are sharing this spectrum and there is interference, then a car that could have prevented the crash would not be able to prevent the crash because someone else is using the spectrum."

Global Automakers, an automotive industry group, is also working to solve the problem of interference by devices that want to use the unlicensed Wi-Fi spectrum currently dedicated to V2V communication.

Cisco Systems has developed a possible way for different industries to share airwaves without such interference, according to a statement from Alliance of Automobile Manufacturers. This is currently in testing as per reports.

## Making daily lives easier

New technologies could also alert drivers when it is time to turn left on a highway exit ramp. Now, this sounds pretty mundane and not really a need for most drivers, but, statistics tell a different story. Unsafe left turns account for more than seven per cent of all car collisions, according to an article in Washington Post.

Car manufacturers like Chevrolet are believed to be planning to use their onboard 4G LTE Wi-Fi and OnStar RemoteLink smartphone app to alert owners if their car is about

to break down. The device can also aid in scheduling a check up at the dealer to prevent the driver from getting stranded on the road.

The IMPACT lab at Arizona State University, the USA, has been developing formal framework for developing criticality-aware software for continuation-passing style (CPS). Such framework is a promising solution to develop criticality-aware future automotive CPS to save numerous lives.

**FPGAs and sensor fusion.** Field programmable gate array (FPGA)-powered systems on module (SoM) technology is used in some cases for sensor-fusion applications in automotive.

"One potential application for this would be to integrate this with the car so that it is able to track your face. It would then be able to detect if you closed your eyes, are drowsy or are texting on your phone. Sensors for lane departure could be connected and processed in real-time through this device. In fact, we are already working with one of our customers in Bengaluru on this same system and have even completed the proof-of-concept," explains Bryan Fletcher, technology director, Global Technical Marketing at Avnet, in an interview with EFY.

Google's self-driving car project

uses a remote-sensing Lidar (laser radar) system to implement driverless technology.

**6D vision technology.** LG says that next-generation camera components will allow a driver to transfer some driving tasks to future intelligent cars by monitoring the driver and environment. The cars would be able to sense sign boards, implement lane-changing and do other driving manoeuvres.

LG will also be allowed to use portions of Mercedes-Benz's 6D Vision technology, which will let a car automatically change speed, depending on route conditions and other situations on the road.

**Autopilot.** Tesla has an impressive self-driving feature called Autopilot, which combines a camera, radar and sonar with real-time traffic data to automatically drive the car. The website explains that, "With Autopilot activated, Model S automatically follows the road, steering around curves and varying its speed to match the flow of traffic. Changing lanes becomes as simple as a tap of the turn signal. When you arrive at your destination, Model S both detects a parking spot and automatically parks itself."

Just like the car-performance upgrade implemented via OTA that we had explained earlier in the article, Autopilot features are progressively enabled over time with software updates. The current software version added automatic emergency braking and blind-spot warning.

Auto manufacturer Ford is partnering with Georgia Institute of Technology for a mobility experiment, Parking Spotter. This project also leverages existing sonar and radar technology already available on Ford vehicles, and uses collected data to map parking information on the go.

The SAM car is an interesting project by a company named Arrow, in which a Corvette car is outfitted to be semi-automatically driven. "Infrared (IR) sensors in the cap worn by the driver provide head position in real-time, which provides data that is used to deduce how much the rotary